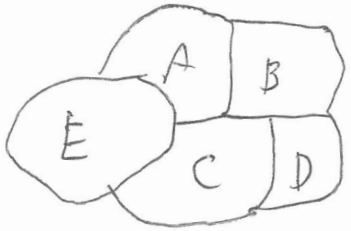
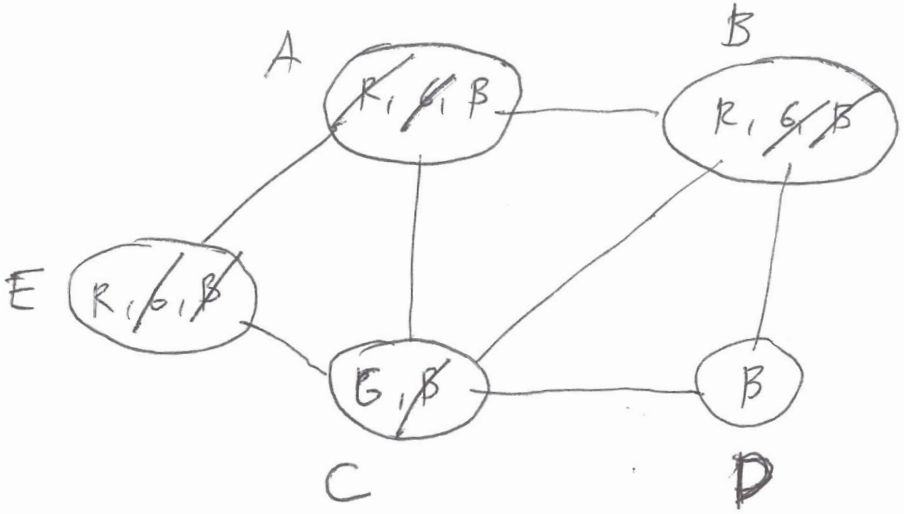
CSP Overview

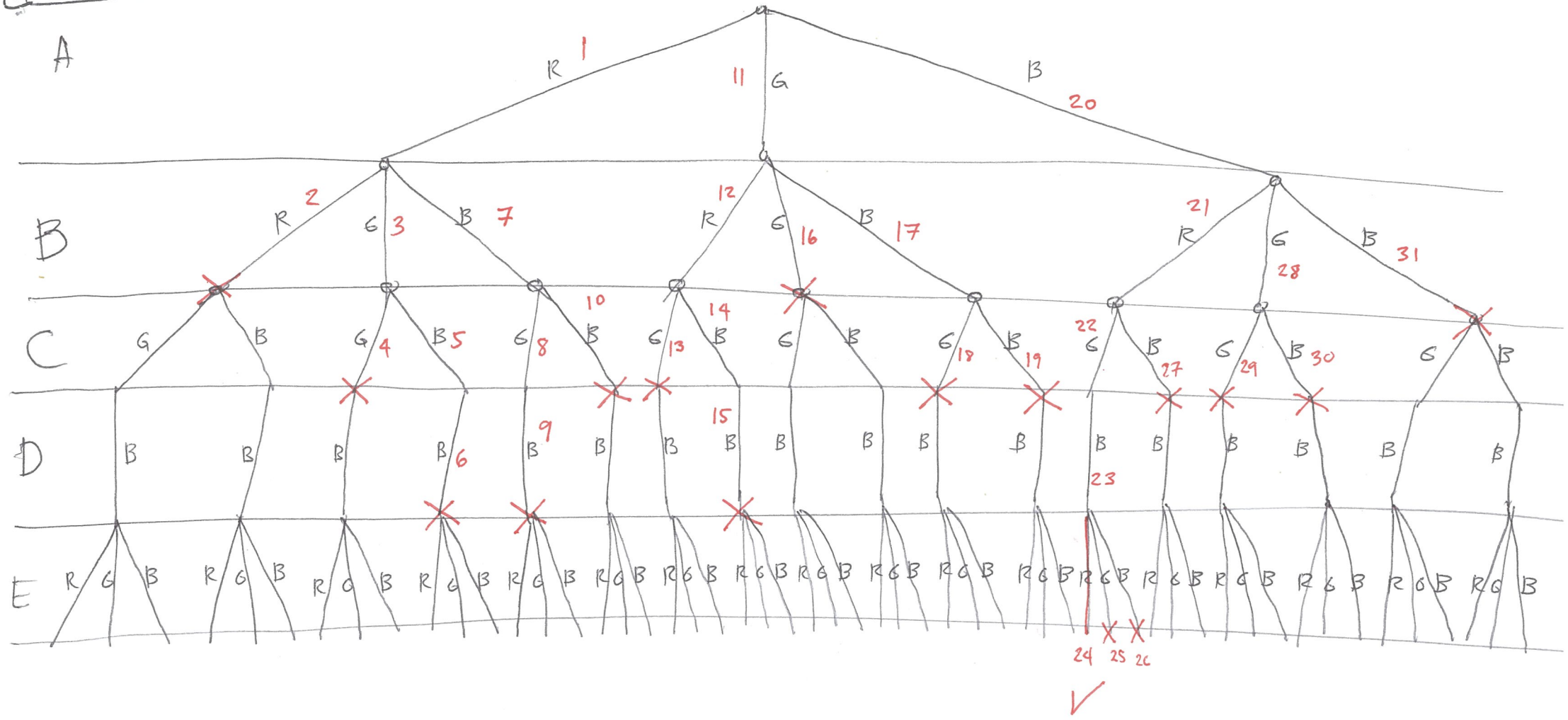
1. F
2. F
3. T
4. T
5. T
6. F
7. F
8. T
9. +

Constraint Propagation



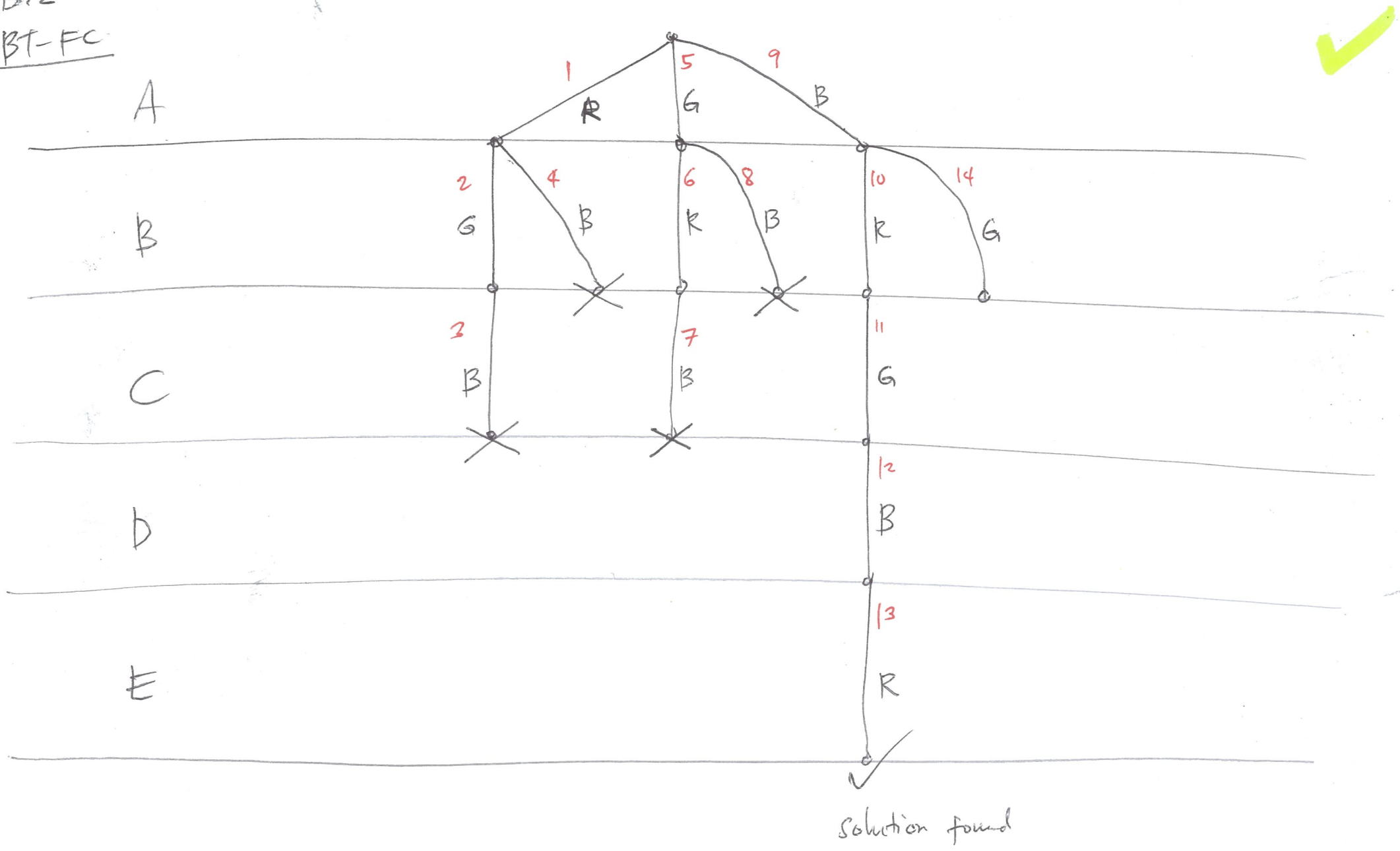


B.1 Pure BT


A
B
C
D
E
R
G
B
1
2
3
4
5
6
7
8
9
10
11
12
13
14
15
16
17
18
19
20
21
22
23
24
25
26
27
28
29
30
31
V

B.2

BT-FC


A
B
C
b
E
1
R
5
G
9
B
2
G
4
B
6
R
8
B
10
R
14
G
3
B
7
B
11
G
12
B
13
R
solution found

BT-FC

1. → A = (R), G, B
   • B = ~~R~~, G, B
   • C = G, B
   D = B
   • E = ~~R~~, G, B

2. A = (R), G, B
   → B = (G), B
   • C = ~~G~~, B
   • D = B
   E = G, B

   A = (R), G, B

3. B = (G), B
   → C = (B)
   • D = ~~B~~ empty
   • E = G, ~~B~~

   ×

Backtrack to B

4. A = (R), G, B
   → B = G, (B)
   • C = G, ~~B~~
   • D = ~~B~~ empty
   E = G, B

   ×

5. Backtrack to A
   → A = (G), B
   • B = R, ~~G~~, B
   • C = ~~G~~, B
   D = B
   • E = R, ~~G~~, B

6. A = (G), B
   → B = (R), B
   • C = B
   • D = B
   E = R, B

7. A = (G), B
   B = (R), B
   → C = (B)
   • D = ~~B~~ empty
   • E = R, ~~B~~

   ×

8. Backtrack to B
   A = (G), B
   → B = (B)
   • C = ~~B~~ } empty
   • D = ~~B~~ }
   E = R, B

   ×

9. Backtrack to A
   → A = (B)
   • B = R, G, ~~B~~
   • C = G, ~~B~~
   D = B
   • E = R, G, ~~B~~

10. A = (B)
    → B = (R), G
    • C = G, ~~B~~
    • D = B
    E = R, G

    A = (B)

11. B = (R), G
    → C = (G)
    • D = B
    • E = R, ~~G~~

    A = (B)

12. B = (R), G
    C = (G)
    → D = (B)
    E = R

13. A = (B)
    B = (R), G
    C = (G)
    D = (B)
    → E = (R)

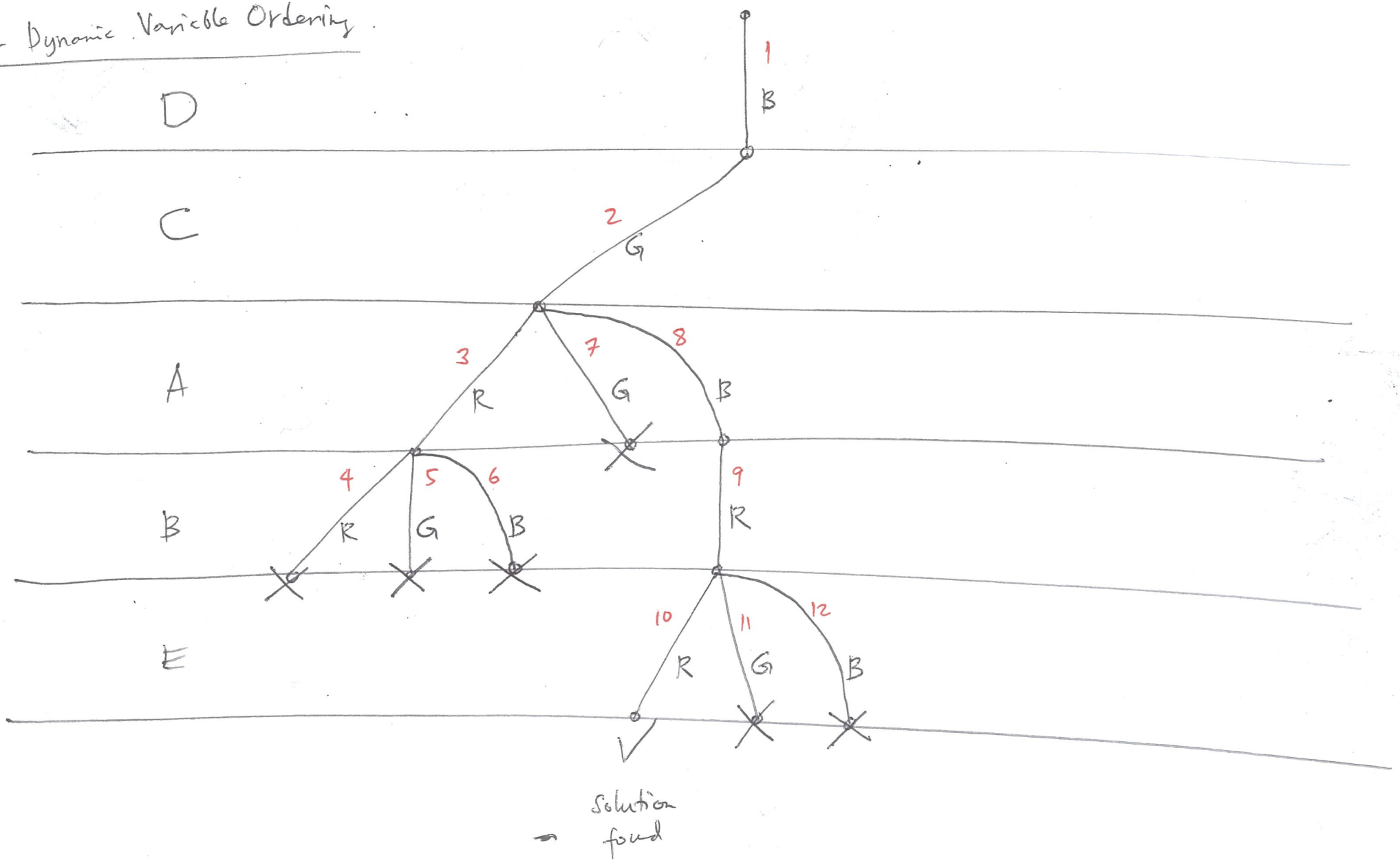
(optional)

14. Backtrack to B
    A = (B)
    → B = (G)
    • C = ~~G~~ empty
    • D = B
    E = R, G

    ×

B.3.

BT - Dynamic Variable Ordering



BT – Dynamic Variable Ordering

0. Most Constrained Variable:

D, C, A, B, E

1. D = (B)

2.

C = G	C = B
A = R, ~~G~~, B	A = R, G, ~~B~~
B = R, ~~G~~, B	B = R, G, ~~B~~
E = R, ~~G~~, B	E = R, G, ~~B~~
rules out 3	rules out 3

(pick the value that rules out the least no. of elements from neighbouring domains)

we have a tie, so pick the one that comes first from the left

C = (G)

3.

A = (R)	A = G	A = B
B = ~~R~~, G, B	B = R, ~~G~~, B	B = R, G, ~~B~~
E = ~~R~~, G, B	E = R, ~~G~~, B	E = R, G, ~~B~~
rules out 2	2	2

4.

B = (R)	B = G	B = B
× constraint violation		

5. B = (G) × constraint violation

6. B = (B) × constraint violation

7. Backtrack to A

A = (G) × constraint violation

8. A = B

9. B = R

10. E = R

11. ~~E~~ = G ×

12. E = B